Hao Ke

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Education

Beijing Jiaotong University, Institute of Information Science *Beijing, China*
M.S. in Information and Communication Engineering *2024 – Present*
• **GPA** 3.4/4.0 **Academic Ranking**: 5/31
• Advisors: Prof. Wei and Prof. Chang

China Three Gorges University *Yichang,Hubei, China*
B.S. in Communication Engineering *2019 – 2023*
• **GPA** 3.0/4.0 (80.0/100)
• **CET4** 500/710 **CET6** 451/710 **IELTS** 6.5/9.0

Publications

Hao Ke, Bo Li, Yakun Chang, Cheng Cheng, Hong Li, Yao Zhao, Shikui Wei. (2026) Glass-Casing-Free 3D Reconstruction. ***IEEE Transactions on Multimedia (TMM)***. 2026.09

- For scenarios involving complex media such as glass that are reflective or semi-transparent.
- A 3D reconstruction method based on implicit representation.

Hao Ke, Bo Li, Yakun Chang, Cheng Cheng, Qinwen Li, Yao Zhao, Shikui Wei. (2026) Consistent 3D Gaussian Scene Update via Voxel-Based Localization. ***(ICRA 2027 Under Review)***. 2026.09

- Proposed voxel-based ray sampling for 3D change localization with reduced sensitivity to primitive distribution.
- A dense depth-guided method for change region localization and 3DGS scene update.

Experience

M.S., Computer Science and Technology, Beijing Jiaotong University 2024 – Present

- Research on 3D reconstruction for challenging scenes such as transparent and reflective environments, consistent 3D Gaussian scene updating and editing, and geometrically consistent feed-forward 3D reconstruction.
- Project: Server Room Fusion Algorithm Technology (Cross-Modal 3D Generation).
  Developed feed-forward 3D geometry initialization and depth adjustment for efficient 3D reconstruction, and continual 3DGS scene updating for routine data-center inspections.
- Project: Based on 3D large-model digital space synthesis technology.
  Develop multimodal 3D reconstruction for semantic-aware, geometrically consistent representations.

Algorithm Engineer(Intern), Automotive and Robotics, Xiaomi Corporation 2026.01 – 2026.10

- Research on Action-Conditional world models for future frame prediction and dynamic scene modeling.
  Developed an action-conditioned world model with autoregressive depth and scheduled latent-noise training for object-level geometric accuracy and robust future frame prediction.
- Research on VLM-based 3D scene representation through depth estimation and 3D occupancy prediction.
- Project: Leveraged V-JEPA 2.1 for spatiotemporal representation learning from multi-view driving scenes.

Competitions and Patents

- Patent: A 3D reconstruction method for transparent casing removal. 202610480833.5 2026
- World Models Track of the GigaBrain Challenge 2026 CVPR Workshop. (Achieved 3rd place) 2026
- The 5th National Undergraduate Embedded Chip and System Design Competition - FPGA Innovation Design Competition. (First Prize) 2021

Awards

- Graduate Academic Scholarship, Beijing Jiaotong University
- Outstanding Graduate, China Three Gorges University